Project Proposal: Space-Time Travel Planner

I. Project Overview

- **Project Name:** Space-Time Travel Planner
- **Proponent:** Lyan Ezekiel P. Tolentino (Section 8-Camia)
- **Subject / Domain:** Computer Science & Interplanetary Physics Simulation
- **Academic Institution:** Philippine Science High School - Eastern Visayas Campus

II. Problem Statement

Albert Einstein's Theory of General Relativity established that massive celestial bodies warp spacetime, dictating that gravitational fields and relative velocities directly influence physical metrics such as weight and temporal progression (Einstein, 1915; Thorne, 1994). When exploring interplanetary transit across different bodies in our solar system—such as Mars, Jupiter, and Venus—varying gravitational constants (\text{Mars} = 0.38g, \text{Jupiter} = 2.34g, \text{Venus} = 0.90g) drastically alter a traveler's effective weight, while extended orbital travel durations shift chronological age relative to Earth years.

Despite these fundamental principles of planetary physics, students and casual space enthusiasts often struggle to visualize or compute these cosmic differentials manually. Complex formulas involving transit windows, calendar increments, and planetary mass ratios can feel abstract and disconnected from practical application. This project addresses the gap by translating theoretical astrophysics into an accessible, interactive digital simulation tool that computes real-time interplanetary weight adjustments, travel durations, arrival dates, and age scaling dynamically based on user input.

III. Project Objectives

General Objective

To design, develop, and test an interactive software program that simulates interplanetary travel logistics, calculating planetary weight variations, transit arrival schedules, and adjusted ages based on celestial gravitational metrics.

Specific Objectives (SMART Framework)

- **Specific (S):** Program a terminal-based application using Python that accepts user weight and age inputs, provides choices across three distinct planetary destinations (Mars, Jupiter, and Venus), and applies specific gravitational multipliers and transit durations.
- **Measurable (M):** Accurately compute and display precise future arrival dates using Python's datetime module, alongside exact planetary weights (rounded to 2 decimal places) and adjusted ages (rounded to 1 decimal place).

- **Achievable (A):** Utilize fundamental programming constructs such as conditional selection structures (if-elif-else), arithmetic operations, and date-time arithmetic matching Grade 8 Computer Science coursework.
- **Relevant (R):** Bridge the intersection of Earth science and computer science by turning abstract celestial mechanics and gravitational physics into a tangible, user-friendly utility.
- **Time-bound (T):** Complete the code implementation, pseudocode, flowchart modeling, and proposal documentation within the prescribed academic quarter timeline.

IV. Planned Features

- **Interactive Destination Selection:** Allows user choice between Mars, Jupiter, and Venus with robust input validation.
- **Gravitational Weight Scaling:** Computes dynamic relative body weight using destination-specific surface gravity constants.
- **Arrival Schedule Calculation:** Computes calendar arrival date dynamically using system departure date and travel transit offsets.
- **Chronological Age Progression:** Computes user age upon arrival based on mission duration.

V. Planned Inputs and Outputs

Feature	Input Data	Expected Output
Weight Scaling	User Earth Weight (numerical, positive float)	Effective Weight on target planet (rounded to 2 decimal places)
Age Progression	User Current Age (numerical, positive float/int)	Chronological Age upon landing (rounded to 1 decimal place)
Destination Setup	Destination Choice (option 1, 2, or 3)	Selected Destination Name, Surface Gravity (g), Transit Duration (days)
Arrival Calculator	System Current Date (datetime.now())	Formatted Arrival Date string (Month Day, Year)

VI. Logic Plan

Pseudocode

```
START
  PRINT "=== SPACE-TIME TRAVEL PLANNER ==="

  INPUT earth_weight
  WHILE earth_weight <= 0
    PRINT "Invalid input. Weight must be positive."
    INPUT earth_weight
  ENDWHILE
```

```
  INPUT earth_age
  WHILE earth_age <= 0
    PRINT "Invalid input. Age must be positive."
    INPUT earth_age
  ENDWHILE

  PRINT "Select Destination:"
  PRINT "1. Mars"
  PRINT "2. Jupiter"
  PRINT "3. Venus"
  INPUT destination_choice

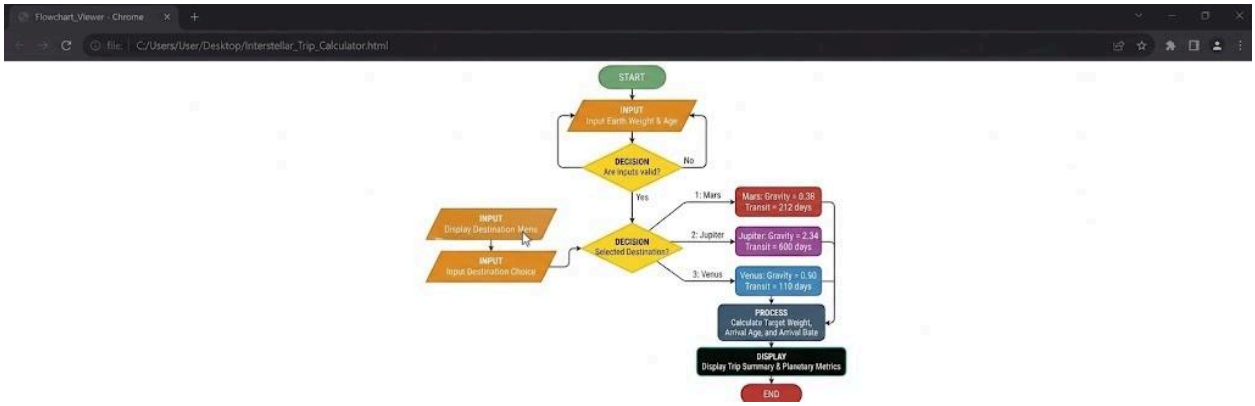
  IF destination_choice == 1 THEN
    destination = "Mars"
    gravity_mult = 0.38
    transit_days = 212
  ELSE IF destination_choice == 2 THEN
    destination = "Jupiter"
    gravity_mult = 2.34
    transit_days = 600
  ELSE IF destination_choice == 3 THEN
    destination = "Venus"
    gravity_mult = 0.90
    transit_days = 110
  ELSE
    PRINT "Invalid choice selected."
    EXIT
  ENDIF

  target_weight = earth_weight * gravity_mult
  arrival_age = earth_age + (transit_days / 365.25)
  departure_date = CURRENT_DATE()
  arrival_date = departure_date + transit_days

  PRINT "=== TRIP SUMMARY ==="
  PRINT "Destination: " + destination
  PRINT "Departure Date: " + departure_date
  PRINT "Travel Duration: " + transit_days + " days"
  PRINT "Estimated Arrival: " + arrival_date

  PRINT "=== PLANETARY METRICS ==="
  PRINT "Weight on " + destination + ": " + target_weight + " lbs"
  PRINT "Age on " + destination + ": " + arrival_age + " years"
END
```

Flowchart


Flowchart_Viewer - Chrome
file:///C:/Users/User/Desktop/Interstellar_Trip_Calculator.html
START
INPUT
Input Earth Weight & Age
DECISION
Are inputs valid?
No
Yes
INPUT
Display Destination Menu
INPUT
Input Destination Choice
DECISION
Selected Destination?
1: Mars
Mars: Gravity = 0.38
Transit = 212 days
2: Jupiter
Jupiter: Gravity = 2.34
Transit = 600 days
3: Venus
Venus: Gravity = 0.90
Transit = 110 days
PROCESS
Calculate Target Weight,
Arrival Age, and Arrival Date
DISPLAY
Display Trip Summary & Planetary Metrics
END